Task 2 - Attitude and Navigation Reference System

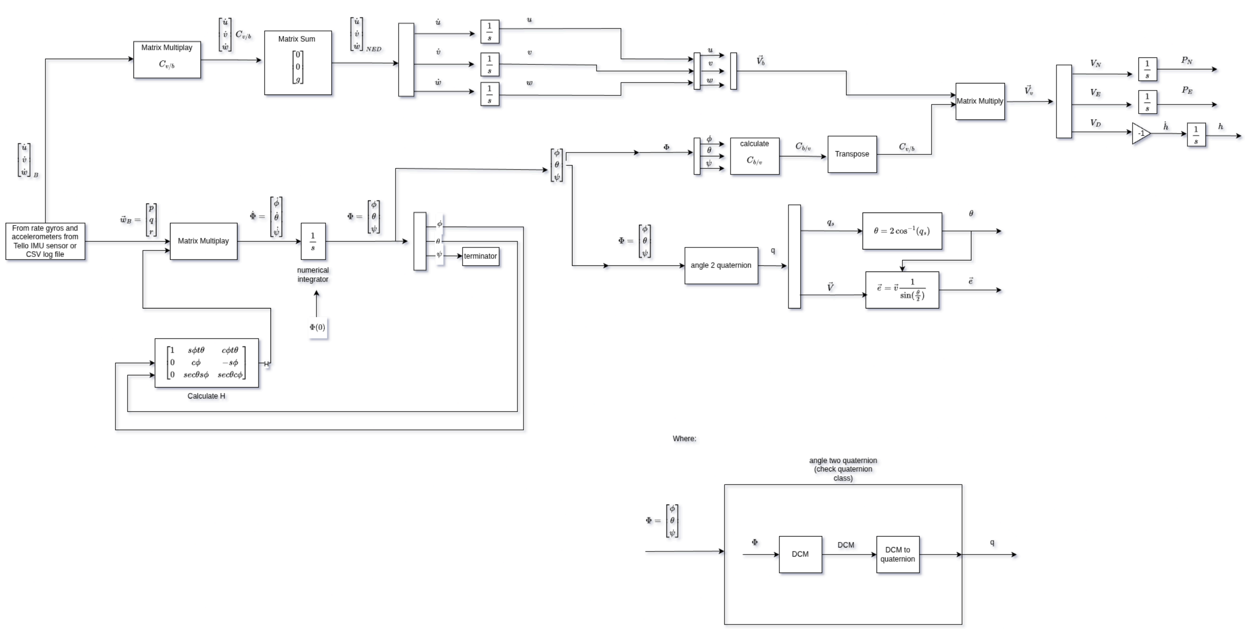
Objective

The goal of this assignment is to implement a **basic inertial navigation and attitude visualization tool** using IMU measurements.

Students will implement the **attitude kinematics**, propagate the aircraft orientation using angular rate measurements, and compute the aircraft **velocity and position in the NED frame**.

Finally, you will visualize the aircraft state through **plots and a graphical interface**.

Implementation Diagram



Task

You will create a program (Python is recommended, but any language is allowed) that:

1. **Implement a attitude heading reference and navigation system where:**
    a. The attitude of aircraft is represented during mission profile.
    b. The position in NED frame is represented during mission profile.
    c. You need to use the raw IMU data from tello drone specified in tello_imu.csv file where the following mission profile is represented:

Time	**Maneuver**	**What students will observe**
0–15 s	Straight & level	stable attitude
15–30 s	Coordinated turn	yaw rate + lateral accel
30–40 s	Climb	positive pitch rate
40–50 s	Descent	negative pitch rate
50–60 s	Level again	stabilization

4. **Output Results**
    a. Graphical Interface: A simple GUI or plots, you are free to use your creativity to create the better interface, but must have at least the following items:
        i. Plots mapped from the data processing showing the 3D or 2D trajectory and the mission profile attitude phases.
        ii. Plots Euler angles during mission profile.
5. Conclusions
    a. There is a fundamental flaw in this approach. Explain what it is and how to correct it. You can check the real trajectory in tello_ground_truth.csv file

Deliverables

1. Code (well commented) in a github repository, check the following tutorial to upload your code and add the link in the report.

2. Short report (Max 2 pages) explaining:
    a. Log analysis describing the mission profile of the aircraft.
    b. Conclusions about flaw and how to fix it

Guidelines

- Deadline: March 25 16:00
- You can use LLMs (Large Language Models) but you must understand all your code. During midterm, the professor is going to ask you about an random section of your code. If you don't answer with criteria, you will only reach half of the score of the task.
- The maximum number of students are three (No exceptions)
- This homework has a value of 1.2 of "midterm 2 - homework" activity.

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